



FEDERAL UNIVERSITY OF LAFIA
FACULTY OF SCIENCE
DEPARTMENT OF ZOOLOGY

SECOND SEMESTER EXAMINATION 2020/2021 ACADEMIC SESSION
BLY 122- GENERAL BIOLOGY II PRACTICAL (1 CREDIT UNITS)

Instruction: Attempt All Questions

Time Allowed: 1 hour



Figure 1: Specimen A



Figure 2: Specimen B

Use the figures above to answer questions 1-6

1. Identify Specimen A above with reasons.
2. What type of circulatory system does specimen A have?
3. Mention the structure found in segment 17 and 18 of specimen A.
4. Identify specimen B
5. Label the parts i - iv of specimen B
6. What is the stain used for the identification of specimen B?
7. Specimen C is a leaf that was plucked and placed in a dark cupboard for seven days while specimen D was freshly plucked leaf from the same plant and boiled in water for a period of six (6) minutes. Thereafter, each specimen (C and D) was dipped into 70% alcohol and then placed in a beaker containing hot water. State what happens when specimens C and D respectively are placed on a white tile and a few drops of iodine solution poured on each.



Using the above diagram, answer question 16-18

16. The parts of the flower represented by P and Q are
 (a). Petal and stigma (b). Sepals and petals (c). Stigma and anther (d). Sepal and stigma

17. The part that attracts pollinators is
 (a). P & Q (b). R & S (c). S & T (d). S

18. The flower is a complete flower because of the presence of
 (a). P, Q, R, S (b). P, Q, T, & U (c). R, S, T, U (d). P, Q, R, T

19. Each nephridium opens to the outside by means of a nephridiopore.

20. What is the significance of the SORI? The SORI is a cluster of sporangia which contains the spores in plants like ferns & moss. The spores are essential for reproduction. SECTION B

1. State Five (5) general and personal laboratory safety measures in a Biology Laboratory
 2. Give any Two (2) adaptive features of the following organisms to aquatic environment

I. Fish - gills - fins.

II. Water lily - big green leaves - round oval or heart-shaped

III. Frog - Bulging eyes and strong legs.

3. With at least two examples, define the following terms;

I. Parasitism - Tapeworms, fleas

II. Asexual reproduction - Amoeba - bacteria

III. Commensalism - Sharks and Remora fish - bee and flower, bacteria and human

IV. Predation - lions, sharks.

4. i. Mention any two external features of fern and cashew plant
 ii. Write four characteristics of a moss plant

fern - seedless vascular plants
plant body is a sporophyte

5. In a tabular form, showing test, observation and inference; describe the translucent test emulsion test for presence of lipids.

Cashew

6. You are provided with specimen A and B. Using external features only;

i. Differentiate between A and B

ii. Similarities between A and B

① Red morphology
② They are evergreen trees

Credit Unit: 1

Instruction: Answer ALL Questions

SECTION A

1. Ferns are epiphytes due to the part called
(a). Rachis (b). Frond (c). Rhizoid (d). Rhizome
2. Lack of true roots, stems and leaves in plants are attributed to
(a). Non-flowering (b). Flowering (c). Vascular bundles (d). Seedling
3. Prokaryotes include the following except
(a) Bacteria (b). Cyanobacteria (c). Blue-green algae (d). Algae
4. Which of the following taxonomic hierarchy exhibits the highest similarities in population?
(a). Class (b). Species (c). Family (d). Genus
5. A class of plant with root nodules is likely to be
(a). Mango (b). Cowpea (c). Cashew (d). Cassava
6. Lateral line in fish is an adaptation for
(a). Buoyancy (b). Counter shading (c). Detection of vibration (d). Steering
7. Tap-rooted and net-veined leaf plants are monocotyledonous plant
8. An aquatic species with hairs caught by a fisherman is most likely to be
(a). Fish (b). Amphibians (c). Mammals (d). Reptile
9. Amplexus is to fish and amphibians whereas _____ is to reptiles and mammals.
10. Lichen and mycorrhiza are examples of
(a). Parasitism (b). Commensalism (c). Mutualism (d). Ammensalism
11. You must do make sure it's thoroughly clean before leaving the laboratory
12. Safety can be
(a) Mechanical, chemical and biological (b). Mechanical, medical and physical (c). Mechanical, corrosive and biological (d). Chemical, biological and physical
13. One observable difference between spirogyra and bread mould is
(a). Cell wall (b). Green coloration (c). Nucleus (d). Cytoplasm
14. Viewing bacteria with oil immersion lens is to classified as high magnification objective
15. Eggs are formed in reproductive system of animals in ovaries